

ĐẠI HỌC BÁCH KHOA HÀ NỘI  
TRƯỜNG CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG

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## BÁO CÁO BÀI TẬP 9/6

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Github: <https://github.com/quocanh13/IT4060>

# Bài 1: Kết Nối Với FTP Server

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <netdb.h>
#include <sys/socket.h>

#define BUF_SIZE 8192

void recv_response(int sock, char *buf, int size)
{
    int len = recv(sock, buf, size - 1, 0);

    if (len <= 0)
    {
        perror("recv");
        exit(1);
    }

    buf[len] = '\0';
    printf("%s", buf);
}

int create_pasv_socket(int control_sock)
{
    char buf[1024];
    int h1, h2, h3, h4, p1, p2;

    send(control_sock, "PASV\r\n", 6, 0);

    recv_response(control_sock, buf, sizeof(buf));

    if (sscanf(buf, "227 Entering Passive Mode (%d,%d,%d,%d,%d,%d)", &h1, &h2, &h3, &h4, &p1, &p2) != 6)
    {
        printf("PASV parse error\n");
        exit(1);
    }

    char ip[32];
    sprintf(ip, "%d.%d.%d.%d", h1, h2, h3, h4);

    int port = p1 * 256 + p2;

    int data_sock = socket(AF_INET, SOCK_STREAM, 0);

    struct sockaddr_in addr;
    memset(&addr, 0, sizeof(addr));

    addr.sin_family = AF_INET;
    addr.sin_port = htons(port);

    inet_pton(AF_INET, ip, &addr.sin_addr);

    if (connect(data_sock, (struct sockaddr *)&addr, sizeof(addr)) < 0)
    {
        perror("connect data");
        exit(1);
    }

    return data_sock;
}

int main()
{
    int client = socket(AF_INET, SOCK_STREAM, 0);

    if (client < 0)
    {
        perror("socket");
        return 1;
    }

    struct addrinfo hints = {0}, *res;
```

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hints.ai_family = AF_INET;
hints.ai_socktype = SOCK_STREAM;

if (getaddrinfo("lebavui.io.vn", "ftp", &hints, &res) != 0)
{
    perror("getaddrinfo");
    return 1;
}

if (connect(client, res->ai_addr, res->ai_addrlen) < 0)
{
    perror("connect");
    return 1;
}

char buf[BUF_SIZE];

recv_response(client, buf, sizeof(buf));

send(client, "USER user_20235263\r\n", strlen("USER user_20235263\r\n"), 0);
recv_response(client, buf, sizeof(buf));

send(client, "PASS 526313\r\n", strlen("PASS 526313\r\n"), 0);
recv_response(client, buf, sizeof(buf));

if (strncmp(buf, "230", 3) != 0)
{
    printf("Login failed\n");
    return 1;
}

send(client, "TYPE I\r\n", strlen("TYPE I\r\n"), 0);
recv_response(client, buf, sizeof(buf));

int data_sock = create_pasv_socket(client);

send(client, "LIST\r\n", strlen("LIST\r\n"), 0);
recv_response(client, buf, sizeof(buf));

char files[BUF_SIZE];
int total = 0;
int len;

while ((len = recv(data_sock, files + total, sizeof(files) - 1 - total, 0)) > 0)
    total += len;

files[total] = '\0';

close(data_sock);

recv_response(client, buf, sizeof(buf));

printf("%s\n", files);

char question_file[256];

char *p = strstr(files, "question_");

if (p == NULL)
{
    printf("Question file not found\n");
    return 1;
}

sscanf(p, "%255s", question_file);

printf("Question file: %s\n", question_file);

data_sock = create_pasv_socket(client);

char cmd[512];

sprintf(cmd, "RETR %s\r\n", question_file);
send(client, cmd, strlen(cmd), 0);

```

```

recv_response(client, buf, sizeof(buf));

char question[1024];

total = 0;

while ((len = recv(data_sock, question + total, sizeof(question) - 1 - total, 0)) > 0)
    total += len;

question[total] = '\0';

close(data_sock);

recv_response(client, buf, sizeof(buf));

printf("Question content:\n%s\n", question);

int qlen = strlen(question);

while (qlen > 0 && (question[qlen - 1] == '\r' || question[qlen - 1] == '\n'))
    qlen--;

char answer[1024];

for (int i = 0; i < qlen; i++)
    answer[i] = question[qlen - 1 - i];

answer[qlen] = '\0';

printf("Answer content:\n%s\n", answer);

char answer_file[256];

strcpy(answer_file, question_file);
memcpy(answer_file, "answer", 6);

printf("Answer file: %s\n", answer_file);

data_sock = create_pasv_socket(client);

sprintf(cmd, "STOR %s\r\n", answer_file);
send(client, cmd, strlen(cmd), 0);

recv_response(client, buf, sizeof(buf));

send(data_sock, answer, strlen(answer), 0);

shutdown(data_sock, SHUT_WR);
close(data_sock);

recv_response(client, buf, sizeof(buf));

send(client, "QUIT\r\n", strlen("QUIT\r\n"), 0);
recv_response(client, buf, sizeof(buf));

close(client);
freeaddrinfo(res);

return 0;
}

```

The image shows a VS Code editor window with a file explorer on the left and a terminal on the right. The file explorer shows a project named 'ASSIGNMENT [WSL: UBUNTU]' with files like 'c\_cpp\_properties.json', 'launch.json', and '9-6-ex1'. The terminal shows the execution of a C program '9-6-ex1-client.c' which is a client for a server. The program sends a 'STOR' command and receives a directory listing. It then sends a question file and receives an answer file. The output shows the directory listing and the content of the files.

```
9-6 > 9-6-ex1 > C 9-6-ex1-client.c > main()
188
189     data_sock = create_pasv_socket(client);
190
191     sprintf(cmd, "STOR %s\n", answer_file);
192     send(client, cmd, strlen(cmd), 0);
193
194     rcv response(client, buf, sizeof(buf));
195
196     printf("Received response: %s\n", buf);
197
198     return 0;
199 }
```

```
quocanh@LAPTOP-IXFU8026:~/code/assignment/9-6/9-6-ex1$ gcc 9-6-ex1-client.c -o 9-6-ex1-client && ./9-6-ex1-client
228 (vsFTPd 3.0.5)
331 Please specify the password.
230 Login successful.
200 Switching to Binary mode.
227 Entering Passive Mode (103,101,163,226,39,124).
150 Here comes the directory listing.
226 Directory send OK.
-rw-r--r-- 1 1002 1002 100 Jun 15 21:29 answerer_slnc89.txt
-rw-r--r-- 1 1002 1002 101 Jun 05 23:13 question_slnc89.txt

Question file: question_slnc89.txt
227 Entering Passive Mode (103,101,163,226,39,159).
150 Opening BINARY mode data connection for question_slnc89.txt (101 bytes).
226 Transfer complete.
Question content:
0eRj0b1z0ruzTFvMTMhR3w8qvduPUIf4I08nFMKj0qhw7Sypfsintw3AXr8aAaaU7Q3L9hpbBORdQgkPkivHq8T1PEv0L2ZhbX

Answer content:
xbHZZ10vEP1TBqHvIkPk6pD0r0tph9L2007UaaA8rXA3WtnisfpyS7MhQDjKMFn8814f1Upudvqaw3RHMTMvFTzur0z1b0jre0
Answer file: answerer_slnc89.txt
227 Entering Passive Mode (103,101,163,226,40,51).
150 Ok to send data.
226 Transfer complete.
221 Goodbye.
quocanh@LAPTOP-IXFU8026:~/code/assignment/9-6/9-6-ex1$
```

Figure 1: Ảnh Kết Quả